

Foysal Mahamud

BSc Computer Science & Engineering | North South University

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PROFILE

CSE undergraduate at NSU with proven ability to ship production-grade software. Built and deployed a multi-role Django + React grant platform used by a real university committee. Strong in full-stack web, C++ systems, and OOP. Committed to clean code and continuous learning.

TECHNICAL SKILLS

Languages

- Python
- Java
- C / C++
- TypeScript
- JavaScript
- SQL

Frontend

- React 19
- HTML5 / CSS3
- TailwindCSS
- Vite

Backend

- Django 4.2
- REST Framework
- JWT / Celery
- Redis

Databases

- PostgreSQL
- MySQL
- Supabase

Tools

- Git & GitHub
- Docker (basics)
- Railway
- Vercel

Other

- STM32 Embedded C
- OOP & DSA
- Agile / Scrum

EDUCATION

B.Sc. Computer Science & Engineering
North South University

2023 – Present • CGPA: 3.10 / 4.00

Higher Secondary Certificate
Science Group

2020 • GPA: 5.00 / 5.00

Secondary School Certificate
Science Group

2018 • GPA: 5.00 / 5.00

SOFT SKILLS

- Problem Solving
- Team Collaboration
- Technical Communication
- Attention to Detail

PROFESSIONAL SUMMARY

Full-stack developer and CSE undergraduate with a record of shipping production-level software. Built the **CTRG Grant Review Management System** — a Django + React platform with FSM-driven proposal workflows, JWT auth, Celery background tasks, and auto-generated PDF/DOCX reports — deployed on Railway and Vercel. Also experienced in **C++ systems programming**, Java OOP, MySQL, and STM32 embedded systems. Writes clean, maintainable code and owns the full development lifecycle.

PROJECTS

CTRG Grant Review Management System

- [github.com/foysal-mahamud/CSE-299-Junior-Design-Course-CTRG-management](#) [+4 / 5 Stars, CSV/DOCX, Celery, JWT](#) **CSE 299 — Junior Design**
- Designed a multi-role platform (PI / Reviewer / SRC Chair) managing the full grant proposal lifecycle: draft → submit → Stage 1 review → revision → Stage 2 review → final decision.
- Implemented Django FSM state machine with django-fsm-log for enforced, auditable state transitions; all business logic strictly server-side.
- Integrated Celery + Redis for automated deadline reminders; ReportLab & python-docx for downloadable PDF/DOCX review reports.
- Deployed React SPA on Vercel, Django API on Railway, PostgreSQL + file storage on Supabase — zero-downtime production environment.

Olympic Athlete Tracking System

- [github.com/foysal-mahamud/CSE-225-Olympic-Project](#) [+4 / 5 Stars, CSV/DOCX, GitHub Pages](#) **CSE 225 — Data Structures**
- CLI application managing Paris 2024 Olympic athlete records using a custom singly-linked list with full CRUD, pagination, and dual search modes.
- Wrote an RFC 4180-compliant CSV tokeniser to correctly parse 30-column quoted data; discovered and fixed 6 column-mapping bugs in original code.
- Built a companion web dashboard (Chart.js) deployed to GitHub Pages showing medal breakdowns, top athletes, and sport statistics.

Stock Buy & Sell Desktop Application

- [github.com/foysal-mahamud/CSE-311-Stock-App](#) [+4 / 5 Stars, MySQL, JDBC, Swing GUI](#) **CSE 311 — Database Systems**
- Relational schema for stock trading with user auth, live portfolio tracking, and transaction history; JDBC with parameterised queries to prevent SQL injection.

Car Racing Game

- [github.com/foysal-mahamud/CSE-215-Car-Racing-Game](#) [+4 / 5 Stars, OOP, Event Handling](#) **CSE 215 — Programming Language II**
- Interactive racing game with OOP design patterns, real-time keyboard event handling, collision detection, and progressive difficulty scaling.

7-Segment Display Controller

- [github.com/foysal-mahamud/CSE-231-7-Segment-Controller](#) [+4 / 5 Stars, STM32, Embedded C, Combinational Logic](#) **CSE 231 — Digital Logic Design**
- Designed combinational + sequential circuits to drive a 7-segment display via STM32 MCU; programmed in Embedded C with direct register manipulation.

RELEVANT COURSEWORK

- CSE 225 Data Structures & Algorithms** — Linked lists, trees, sorting, hashing, complexity analysis
- CSE 311 Database Systems** — ER modelling, normalisation, SQL, JDBC, transactions
- CSE 327 Software Engineering** — SDLC, Agile/Scrum, sprint planning, Kanban, XP
- CSE 231 Digital Logic Design** — Boolean algebra, combinational/sequential circuits, STM32 lab
- CSE 215 Programming Language II** — Advanced Java OOP, generics, collections, GUI programming